NEOGUARD | ESP32-S3 SYSTEM CONNECTIONS

DRAFT E: direct signal wiring and ground symbols. USB-C battery power design is pending the battery specification.

MAX30102 / 0x57

J2

VCC 1 → V3V3
GND 2
SDA 3 → I2C_SDA_GPIO4 → GPIO4_/_SDA 4
SCL 4 → I2C_SCL_GPIO5 → GPIO5_/_SCL 5
Pin_5 5 — INT / unused

MLX90614 / 0x5A

J3

VCC 1 → V3V3
GND 2
SDA 3
SCL 4

DS1302 RTC

J4

VCC 1 → V3V3
GND 2
CLK 3 → RTC_CLK_GPIO6 → GPIO6_/_CLK 6
DAT 4 → RTC_DAT_GPIO7 → GPIO7_/_DAT 7
CE_/_RST 5 → RTC_CE_GPIO14 → GPIO14_/_CE 14

EXTERNAL RGB LED

J7

RED_A 1 → LED_R_A → R3 330 → LED_R_GPIO16 → GPIO16_/_RED 16
GREEN_A 2 → LED_G_A → R4 330 → LED_G_GPIO17 → GPIO17_/_GREEN 17
BLUE_A 3 → LED_B_A → R5 330 → LED_B_GPIO18 → GPIO18_/_BLUE 18
COMMON_K 4

330 ohms per colour; common cathode to GND.

U1 — ESP32-S3 DEVELOPMENT BOARD

DEV3V3 → 3V3 → 3V3_OUT

GPIO0 0
GPIO1 1
GPIO2 2
GPIO3 3
GPIO13 13
GPIO19 19
GPIO20 20
GPIO35 35
GPIO36 36
GPIO37 37

GPIO38 38
GPIO39 39
GPIO43 43
GPIO44 44
GPIO45 45
GPIO46 46
GPIO47 47
GPIO48 48
EN EN
5V 5V

GPIO8_/_RST 8 → TFT_RST_GPIO8
GPIO9_/_DC 9 → TFT_DC_GPIO9
GPIO10_/_CS 10 → TFT_CS_GPIO10
GPIO11_/_MOSI 11 → TFT_MOSI_GPIO11
GPIO12_/_SCK 12 → TFT_SCK_GPIO12

GPIO21_/_CS 21 → SD_CS_GPIO21
GPIO40_/_SCK 40 → SD_SCK_GPIO40
GPIO41_/_MOSI 41 → SD_MOSI_GPIO41
GPIO42_/_MISO 42 → SD_MISO_GPIO42

GPIO15_/_BUZZ 15 → BUZZ_GPIO15

Logical GPIO interface
External connections
X = unused carrier pin

GND → GND

ST7789 TFT / 240 x 280

J5

RST 5 ← TFT_RST_GPIO8
DC 6 ← TFT_DC_GPIO9
CS 7 ← TFT_CS_GPIO10
MOSI 4 ← TFT_MOSI_GPIO11
SCK 3 ← TFT_SCK_GPIO12
GND 1
V3V3 → VCC 2
TFT_BL → BL_/_BLK 8

SPI MICROSD MODULE

J6

CS 6 ← SD_CS_GPIO21
SCK 3 ← SD_SCK_GPIO40
MOSI 4 ← SD_MOSI_GPIO41
MISO 5 ← SD_MISO_GPIO42
SD_VCC → VCC 1
GND 2

BUZZER + DRIVER

BUZZ_BASE
R6 1k
R7 10k
Q1

D1
BUZZ_VCC → POSITIVE 1
BUZZ_SW ← NEGATIVE 2

J8

Q1 PN2222A: E1 / B2 / C3
D1 1N5819: stripe/cathode toward buzzer +
Active-high driver; buzzer 20 mA maximum.

Power and carrier harness

File: NeoGuard-power.kicad_sch

Selected V3V3 powers the modules; it is chosen on sheet 2. DEV3V3 is the controller output, not a short to external 3V3.

Default screen is the working ST7789 TFT. A different OLED needs its own verified interface. GPIO40-42 must be available on the actual board.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /
File: NeoGuard.kicad_sch

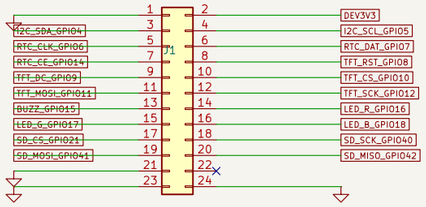
Title: NeoGuard - ESP32-S3 System Connections

Size: A3	Date: 2026-10-29	**Rev: E-POWER**
KiCad E.D.A. 10.0.4		Id: 1/2

NEOGUARD | POWER AND CARRIER HARNESS

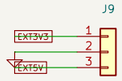
This sheet preserves the real carrier connector pin numbers and the selectable supply topology.

J1 / CONTROLLER CABLE



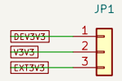
J1 is a cable connector, not a devboard socket.
Map each pin to the GPIO label on U1 / sheet 1.
Pin 2 is DEV3V3 from the ESP32 board.
Pins 1, 21, 23, 24 are ground; pin22 is unused.

INPUT SUPPLIES



J9 accepts already regulated supplies:
1 = external 3V3; 2 = GND; 3 = external 5V.
No onboard regulator or reverse-polarity protection.

JP1 / CARRIER 3V3 SOURCE



One shunt only:
1–2 = DEV3V3 -> selected V3V3
2–3 = EXT3V3 -> selected V3V3
Never fit both positions.

JP2 / SD MODULE SUPPLY

1–2 = V3V3; 2–3 = EXT5V.
One shunt only. Match the actual
SD breakout supply rating.

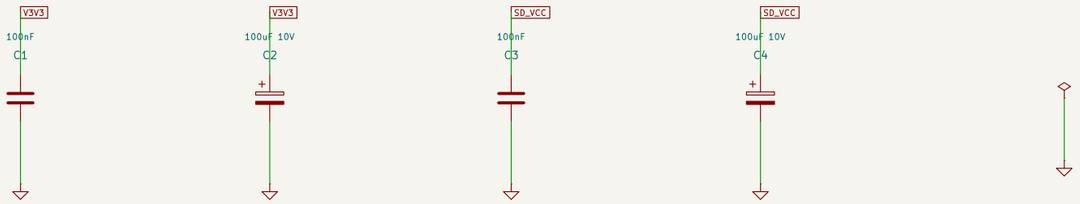
JP3 / BUZZER SUPPLY

1–2 = V3V3; 2–3 = EXT5V.
One shunt only. Match the buzzer
rating; this driver is for <=20 mA.

JP4 / TFT BACKLIGHT

Default OPEN / DNP.
Connect only for a verified 3V3-compatible
BL input with suitable driver/current limiting.

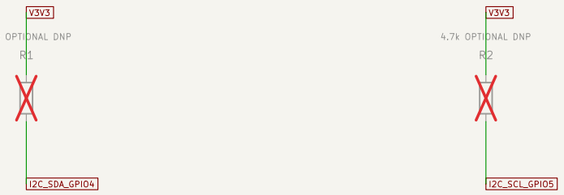
DECOUPLING / BULK STORAGE



Both copper layers use GND pours with thermals and stitching vias. Keep controller and modules powered together; signal isolation is not provided.

All carrier interfaces use 3.3V logic. Confirm breakout voltages, physical connector order, purchased-part pinouts and current budget before fabrication.

OPTIONAL I2C PULL-UPS



R1/R2 are DNP if module pull-ups already exist.
I2C pull-ups go to selected V3V3, never 5V.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /Power and carrier harness/
File: NeoGuard-power.kicad_sch

Title: NeoGuard – Power and Carrier Harness

Size: A3	Date: 2026-10-29	**Rev: E-POWER**
KiCad E.D.A. 10.0.4		Id: 2/2